

What is a PCBA?

Printed circuit board assembly (PCBA) is the board obtained after applying solder paste to the PCB and then mounting various components like resistors, integrated circuits (ICs), capacitors and transformers depending on the application and desired characteristics of the board. A PCBA usually undergoes reflow furnace heating to establish mechanical connection between the PCB and components.

Consumer electronics, LED lighting, automotive electronics and industrial electronics are all major drivers for the PCB industry.

The current US\$ 2.38-billion market represents demand for all types of PCBs, including bare-board as well as populated PCBs. Only 35 per cent of this demand is met by local manufacturers. Demand for bare-board PCBs is worth US\$ 1.2 billion. Only 30 per cent of this demand is met by local PCB manufacturers. Thus this is the perfect time to start PCB assembly (PCBA) business.

Business setup

According to the size of operation, PCBA manufacturing units can be categorised as follows:

1. Large-volume manufacturing
2. Small-volume manufacturing
3. Small-volume fast-delivery manufacturing
4. Small-volume complex-design PCB manufacturing

Requirements of space, machinery, utilities like water and electricity, etc vary for all these categories.

Raw materials. The major raw materials required include solder, flux, chemicals, adhesive, wires and PCB. The PCBA process always starts with the base, which is the most basic unit of a PCB. It consists of several layers, each of which plays a significant role in the functionality of the final PCB. These alternating layers include:

1. **Substrate.** This is the base material that imparts rigidity to the PCB.
2. **Copper.** A thin layer of conductive copper foil is added to functional sides of the PCB. This is the layer of copper traces. In single-side PCBs, copper is applied on only one side. Double-side PCBs have copper on both sides.
3. **Solder mask.** On top of the

copper layer is the solder mask, which gives each PCB its characteristic green colour. It insulates copper traces from unintentionally contacting other conductive materials, thus preventing shorting. The solder keeps everything in its place. Holes in the solder mask are where solder is applied to attach components to the board. Solder mask is a vital step for smooth manufacturing of PCBA since it stops soldering on unwanted parts with shorts avoided.

4. **Silkscreen.** A white silkscreen is the final layer on the PCB board. It adds labels to the PCB in the form of characters and symbols. This helps to indicate the placement of all components on the board.

These materials and components remain largely the same across all PCBs, with the exception of the substrate. The substrate material of a PCB changes according to the finished product requirements such as low cost and bendability.

Machinery and tools. The must-have PCB assembly machineries include:

1. Heat-wave soldering machine
2. Oscilloscope (50MHz)
3. LCR-Q meter
4. Function generator
5. Power supply (0-30V, 3A)
6. 4½-digit digital multimeter

7. Analogue multimeter
8. Magnifying glass fitted with tubelight
9. Temperature-controlled soldering station
10. Soldering iron (25W)
11. Soldering iron (65W)
12. Variac
13. Rheostat

Additional machineries, equipment and tools may be required depending on the nature of the final product, scale of operation, quality compliance requirements, etc. These include but are not limited to:

1. Torque-controlled screwdrivers (electric and pneumatic) to tighten screws and to speed up the production work. Various models are available in electric (AC and DC) and pneumatic categories.
2. Liquid dispensers to dispense glue, epoxy, solder paste, etc. Different kinds of dispensers are available, such as general dispensers, robotic dispensers, rotary dispensers, valves and valve controllers.
3. BGA and other repair or rework machinery
4. Tools such as wire and cable strippers, electronic pliers and screwdrivers, crimping tools, thermal wire strippers, heat guns, cable cutters, test and measuring instruments for power utility, and LAN testers.
5. Adhesives and cleaning products, cases and tool kits, ESD-control products, power tools, test instruments for field and portable use, wire-wrap tools
6. Microscopes, magnifying lamps, magnifiers, professional tool kits, pliers, cutters, precision screwdrivers and tweezers, alignment

Compliance checklist

1. Business registration certificate as per Ministry of Corporate Affairs, government of India regulations
2. Trade licence from the local municipal authority
3. Udyog Aadhaar MSME Registration
4. No-objection certificate (NOC) from Pollution Control Board and effluent treatment plant as per the norms of the Board
5. ISO certification for better customer acceptance. For example, ISO9000 defines standards for quality management systems and ISO14001 defines standards for environment management system, ensuring acceptability at the international level
6. Trademark registration to secure company brand
7. GST registration and other yearly compliance liabilities